

Ujaan Mukherjee

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B.Tech CSE student with a strong command of C++, low-level programming, and systems, comfortable working close to the metal across Linux internals, memory and performance, and systems security. Also build across the stack, from real-time embedded / IoT to ML and edge backends, with hands-on work in RAG, agentic AI, and LLM pipelines.

TECHNICAL SKILLS

Languages	C++, Python, Bash, JavaScript, Java
Systems & Embedded	ESP32, ESP8266, Arduino, real-time data streaming, Linux (WSL), Docker, Git, Oracle VirtualBox, OpenCV
Backend & Edge	Node.js, Express.js, Hono, Cloudflare Workers, REST API design
AI / ML	RAG, LangChain, CLIP, LLM fine-tuning, transformer models, NLP, prompt engineering

EXPERIENCE

AI Engineering Intern | Agenscience.ai June 2026 - August 2026

- Built 3 AI-powered prototypes for a decision-intelligence platform and ran the full ML lifecycle on client datasets
- Designed and tested multi-agent LLM pipelines with tool-use and agentic orchestration; presented across 5+ team sessions

Technical Director | JAAGO NGO, Kolkata 2026 - Present

- Own all technical operations; built and shipped the organisation's website solo, eliminating third-party vendor costs

Physics Virtual Labs Intern | IEM IIFR, Kolkata May 2026 - August 2026

- Developed and debugged virtual physics lab simulations, improving physical accuracy by ~15%

PROJECTS

Solan | open-source C++ project github.com/KireinaR/solan
C++, systems programming

- Designed and built Solan from scratch in modern C++, focusing on clean architecture, predictable performance, and careful memory management
- Maintain it as an open-source project with documentation and a live site, handling issues and iterating on the public API
- Used the project to go deep on low-level and systems concepts, RAII, data layout, and profiling for performance

Heat Seeker: Real-Time Embedded Thermal Tracking

ESP32, MLX90640, C++, Python, OpenCV

- Built a real-time thermal tracker on ESP32 + MLX90640 under tight on-device memory and timing constraints, achieving sub-100 ms end-to-end latency
- Delivered heat-signature detection and tracking at 90%+ accuracy with a Python / OpenCV host pipeline
- Designed a fully self-contained, plug-and-play deployment in under 5 minutes with zero cloud dependency

Argus: Semantic Face Search System

Python, CLIP, NumPy, GPU

- Engineered a CLIP-based semantic search system over 14,000+ images, cutting query time ~70%
- Optimised the embedding pipeline, batching and memory / I-O layout tuning, to reduce disk I/O ~60%

Global Air Bases REST API

Hono, Cloudflare Workers, REST

- Deployed a global edge API (2,000+ air bases, 50+ countries) serving responses in under 10 ms at zero server cost
- Architected a scalable geospatial schema with query filtering

YDSE Simulator: Interactive Physics Web App

JavaScript, Vercel

- Implemented wave interference and diffraction from first principles, sustaining 60 FPS across major browsers

EDUCATION

Institute of Engineering and Management (IEM), Salt Lake, Kolkata	2025 - Present
B.Tech, Computer Science and Engineering	

Don Bosco School, Liluah (CISCE)	2023 - 2025
Class 12: 89.5% (2025); Class 10: 93% (2023)	

CERTIFICATIONS

IBM	AI Agents with RAG and LangChain; Advanced Fine-Tuning for LLMs; Fine-Tuning Transformers; Foundation Models for NLP
Google Cloud	Introduction to Generative AI
Duke University	How LLMs Work; Python Scripting